Kristian **Schwethelm**

Ph.D. Student · Large Language Models · Technical University of Munich

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Summary

Ph.D. student working across the LLM stack, from pretraining and reinforcement learning to efficient inference and practical applications. Projects include looped LM scaling laws and efficient inference. Publications include several first-author papers, among them a NeurIPS Spotlight and an ICLR paper. Interested in full-time positions and internships in LLM research and engineering.

Skills

- **Programming and tools:** Python, C++ (earlier work), coding agents
- **ML systems:** PyTorch, JAX, Hugging Face Transformers, vLLM, SGLang, Slurm, multi-node training
- **Research:** LLM pretraining and post-training (SFT, GRPO), efficient inference, agent evaluation, computer vision, differential privacy
- **Languages:** German (native), English (fluent), Slovenian (fluent)

Work Experience

Technical University of Munich — *Munich, Germany*

Ph.D. Student — Large Language Models — *Oct. 2023 – Present*

- Developed scaling laws for looped LMs to measure how reusing transformer layers affects model quality at the same compute budget.
- Designed and evaluated an inference method for looped LMs that increased serving throughput by up to 53%.
- Built an LLM-assisted text extraction pipeline for hospital notes, achieving 92–99% extraction accuracy.
- Developed an agentic clinical diagnosis benchmark to evaluate LLM agents.
- Teaching: Supervised M.Sc. theses, taught graduate seminars, and trained colleagues to use coding agents.
- Earlier research on differential privacy and privacy auditing.

University of Hildesheim (Data Science Group) — *Hildesheim, Germany*

Student Research Assistant — Deep Learning — *Nov. 2021 – Sep. 2023*

- Developed hyperbolic neural networks for computer vision; published at ICLR 2024.

Robert Bosch GmbH (Corporate Research) — *Hildesheim, Germany*

Working Student — Multi-View Scene Analysis — *Apr. 2021 – Oct. 2021*

- Collaborated with research engineers for multi-camera calibration and 3D object reconstruction algorithms for automotive perception.

Robert Bosch GmbH — *Hildesheim, Germany*

Dual Study Student — *Sep. 2016 – Mar. 2021*

- Fully funded dual study program with industry experience across several R&D and software engineering departments.

Selected Publications

- How Much Is One Recurrence Worth? Iso-Depth Scaling Laws for Looped Language Models, NeurIPS 2026 Spotlight
- Depth-adaptive Inference of Looped Language Models via Continuous Depth Batching, preprint 2026
- Visual Privacy Auditing with Diffusion Models, TMLR 2025
- Fully Hyperbolic Convolutional Neural Networks for Computer Vision, ICLR 2024

Honors & Awards

2025	**Top Reviewer**, NeurIPS 2025
2024	**Best Graduate Award**, M.Sc. Applied Computer Science
2016–2023	**Scholarships**, Deutschlandstipendium (2021, 2022, 2023), State Scholarship Lower Saxony (2019), Full Industry Sponsorship Robert Bosch GmbH (2016–2021)

Education

M.Sc. Applied Computer Science — Grade: 1.0 (Outstanding)

University of Hildesheim, Germany — *Apr. 2021 – Feb. 2023*

- Specialization: Machine Learning and Deep Learning. Thesis: *Hyperbolic Variational Autoencoder for Image Generation.*

B.Eng. Electrical and Information Engineering — Grade: 1.09 (Outstanding)

Ostfalia University of Applied Sciences, Wolfenbüttel, Germany — *Sep. 2016 – Mar. 2021*

- Thesis: *Camera-Based Object Detection Using Transformer Architectures.*